

Fundamentals of theory of computation I.

Lecture 8

Krisztián Tichler
ktichler@inf.elte.hu

CONTEXT-FREE GRAMMARS

From now on we consider context-free grammars.

Reminder: A grammar $G = (N, \Sigma, P, S)$ is called a context-free grammar (type 2 grammar) if each rule of P is of the form $A \rightarrow u$, where $A \in N, u \in (\Sigma \cup N)^*$.

Active and reachable nonterminals

Definition

A nonterminal $X \in N$ of a context-free grammar $G = (N, \Sigma, P, S)$ is called **active** if there is a word $w \in \Sigma^*$ such that $X \Rightarrow_G^* w$ holds. Otherwise it is called **inactive**.

Active nonterminal: at least one terminal word is derivable from it.

Definition

A nonterminal $Y \in N$ of a context-free grammar $G = (N, \Sigma, P, S)$ is called **reachable** if there are words $v_1, v_2 \in (N \cup \Sigma)^*$ such that $S \Rightarrow_G^* v_1 Y v_2$ holds. Otherwise it is called **unreachable**.

Reachable nonterminal: it appears in at least one sentential form (i.e., in a word derivable from the initial symbol).

Definition

A context-free grammar is called **reduced**, if all of its nonterminals are active and reachable.

Determining the set of active nonterminals

Let $G = (N, \Sigma, P, S)$ be a context-free grammar.

$$A_1 := \{X \mid X \rightarrow u \in P, u \in \Sigma^*\},$$

$$A_{i+1} := A_i \cup \{X \mid X \rightarrow w \in P, w \in (\Sigma \cup A_i)^*\}, \quad i = 1, 2, \dots$$

Obviously, $A_1 \subseteq A_2 \subseteq \dots \subseteq N$.

So, $\exists k \in \mathbb{N}$, such that $A_k = A_{k+1}$. Then $A_\ell = A_k$ holds $\forall \ell \geq k$.

$A := A_k$.

Theorem

Set A defined above is the set of active nonterminals of grammar G .

Determining the set of active nonterminals

Proof: Let $X \in A$ be arbitrary, then we have $X \in A_k$.

Claim: For all $0 \leq i \leq k$ there exists $w \in (\Sigma \cup A_{k-i})^*$ with the property $X \Rightarrow_G^* w$. ($A_0 := \emptyset$).

We prove the claim by induction on i .

The claim holds for $i = 0$ since we have $X \Rightarrow^* X$ and $X \in A_k \subseteq (\Sigma \cup A_{k-0})^*$.

Assume, that there exists $w \in (\Sigma \cup A_{k-i})^*$ with $X \Rightarrow_G^* w$. Then by the definition of A_{k-i} a word from $(\Sigma \cup A_{k-(i+1)})^*$ can be derived in one step from every nonterminal of w . Let us derive such a word from every nonterminal of w so there exists a word $w' \in (\Sigma \cup A_{k-(i+1)})^*$ with the property $X \Rightarrow_G^* w \Rightarrow_G^* w'$, proving the claim.

Applying the claim for $i = k$ we have that all nonterminals of A are active.

Determining the set of active nonterminals

Let X be an active nonterminal of G . Then there exists a derivation $X = w_n \Rightarrow w_{n-1} \Rightarrow \cdots \Rightarrow w_1 \Rightarrow w$ in G for $n \geq 1$ and some $w \in \Sigma^*$.

Claim: $w_i \in (A_i \cup \Sigma)^*$ ($1 \leq i \leq n$).

We prove the claim by induction on i .

For $i = 1$: the last step of derivation should rewrite a nonterminal of A_1 .

Assume, that the claim holds for $i(\geq 1)$. The nonterminal rewritten in the derivation step $w_{i+1} \Rightarrow w_i$ is replaced by a subword of $w_i \in (A_i \cup \Sigma)^*$ so this subword is in $(A_i \cup \Sigma)^*$ as well. So, the nonterminal rewritten is from A_{i+1} implying $w_{i+1} \in (A_{i+1} \cup \Sigma)^*$ and proving the claim.

Therefore, $X = w_n \in (A_n \cup \Sigma)^*$, and we have $X \in A_n \subseteq A$.

So, every active nonterminal is in A proving the theorem.

Determining the set of reachable nonterminals

Let $G = (N, \Sigma, P, S)$ be a context-free grammar.

$$R_1 := \{S\},$$

$$R_{i+1} := R_i \cup \{Y \in N \mid X \rightarrow uYw \in P, X \in R_i, u, w \in (N \cup \Sigma)^*\},$$
$$i = 1, 2, \dots$$

Obviously $R_1 \subseteq R_2 \subseteq \dots \subseteq N$.

So, $\exists k \in \mathbb{N}$, such that $R_k = R_{k+1}$. Then $R_\ell = R_k$ holds $\forall \ell \geq k$.

$R := R_k$.

Theorem

Set R defined above is the set of reachable nonterminals of grammar G .

Determining the set of reachable nonterminals

Proof: We prove by induction on i that every nonterminal of R_i is reachable.

This is true for $i = 1$ by definition.

Assume, that the statement holds for i and let $Y \in R_{i+1}$.

Then there exists $X \in R_i$, $u, w \in (N \cup \Sigma)^*$ with $p = X \rightarrow uYw \in P$.

By induction there exists $v_1, v_2 \in (N \cup \Sigma)^*$, with the property

$S \Rightarrow^* v_1 X v_2$. Applying p we get $S \Rightarrow^* v_1 X v_2 \Rightarrow v_1 u Y w v_2$, so Y is reachable.

Determining the set of reachable nonterminals

On the other hand let $S = w_1 \Rightarrow \dots \Rightarrow w_n = uZw$ be a derivation ($n \geq 1$), where $w_i, u, w \in (N \cup \Sigma)^*$ ($1 \leq i \leq n$) and $Z \in N$.

We prove by induction on i that every nonterminal of w_i is in R_i .

The statement is obvious for $i = 1$.

Assume, that the statement holds for i and let Y be a nonterminal of w_{i+1} . If Y appears in w_i , then we have $Y \in R_i \subseteq R_{i+1}$ by induction. If Y does not appear in w_i then a rule $X \rightarrow uYw \in P$ is applied in the derivation step $w_i \Rightarrow w_{i+1}$ where $u, w \in (N \cup \Sigma)^*$.

We have $X \in R_i$ by induction implying $Y \in R_{i+1}$.

So $Z \in R_n \subseteq R$ proving the theorem.

Reduced grammar

Theorem

For every context-free grammar G there exists a reduced context-free grammar G' equivalent to G .

Proof: First, delete all rules of G containing at least one inactive nonterminal (G''), then delete all rules containing unreachable nonterminals (G').

Obviously, G' is equivalent to G and all of its nonterminals are reachable. Let X be a nonterminal of G' . X is active G'' , so there is a terminal word derivable from X in G'' . Since X is reachable in G' all the nonterminals involved in that derivation is reachable in G' , so this derivation is still there in G' , proving that X active in G' .

So G' is a reduced grammar equivalent to G .

Remark: The order is important, we should start with deleting the inactives first then the unreachable ones, second.

Reduced grammar

Since all type 3 grammars are of type 2 as well and all the transformations are just deletion of rules we have the following.

Corollary

For every regular grammar G there is an equivalent reduced regular grammar G' .

Reduced grammar

Example

$S \rightarrow A \mid bBD$
 $A \rightarrow AB \mid AC$
 $B \rightarrow \varepsilon \mid a \mid SS$
 $C \rightarrow AS \mid aba$
 $D \rightarrow BB$

Active nonterminals:

$A_1 = \{B, C\}, A_2 = \{B, C, D\},$

$A_3 = A_4 = \{B, C, D, S\},$

so rules containing A can be eliminated.

$S \rightarrow bBD$
 $B \rightarrow \varepsilon \mid a \mid SS$
 $C \rightarrow aba$
 $D \rightarrow BB$

Reachable nonterminals:

$R_1 = \{S\}, R_2 = R_3 = \{S, B, D\},$

so C can be eliminated.

$S \rightarrow bBD$
 $B \rightarrow \varepsilon \mid a \mid SS$
 $D \rightarrow BB$

The result is reduced and equivalent to the original one.

Chomsky normal form

Definition

A grammar $G = (N, \Sigma, P, S)$ is of **Chomsky normal form**, if each rule of P is of one of the following form

- ▶ $S \rightarrow \varepsilon$
- ▶ $A \rightarrow BC$, for some $A, B, C \in N$; furthermore $B, C \neq S$, if $S \rightarrow \varepsilon \in P$.
- ▶ $A \rightarrow a$, for some $A \in N, a \in \Sigma$.

Theorem

For every context-free grammar G there is an equivalent grammar G' in Chomsky normal form.

Proof: A sequence of transformations is performed on G . Each step transforms the grammar to an equivalent grammar.

Chomsky normal form

Step 1 Eliminating rules with nonsolitary terminals

For all $a \in \Sigma$ we do the following. Let $\bar{a}$ be a new nonterminal. Replace all occurrences of a by $\bar{a}$. Add the rule $\bar{a} \rightarrow a$ to P .

The resulted grammar is equivalent to G (already proved on Lecture 2).

Exception: If $A \rightarrow a \in P$, then no replacement of a is needed. This rule is allowed in a Chomsky normal form.

Chomsky normal form

Step 2 Length reduction

Consider a too lengthy rule $X \rightarrow Y_1 Y_2 \cdots Y_k$, $k \geq 3$.
This rule is replaced by the set of rules

$$\{X \rightarrow Y_1 Z_1, Z_1 \rightarrow Y_2 Z_2, \dots, Z_{k-2} \rightarrow Y_{k-1} Y_k\}$$

where $Z_1, \dots, Z_{k-2}$ are new nonterminals.

Obviously all application of the rule $X \rightarrow Y_1 Y_2 \cdots Y_k$ can be replaced by the application of the corresponding set of short rules. On the other hand, any derivation containing Z_i in the grammar after the transformation should contain nonterminals all the nonterminals $Z_1, \dots, Z_{k-2}$ in this order. It can be proved, that if the derivation step bringing in Z_i is not followed by the application of the rule for Z_i , then the order of derivation steps can be modified to do so without changing the word derived. So applications of the new rules can be grouped in such a way that they correspond to applications of the original rules.

Chomsky normal form

Step 3 Eliminating ε -rules

Let us define the sets $U_i \subseteq N$ as follows.

$$U_1 := \{X \mid X \rightarrow \varepsilon \in P\},$$

$$U_{i+1} := U_i \cup \{X \mid X \rightarrow u \in P \text{ and } u \in U_i^*\}. \quad i \geq 1$$

Obviously $U_1 \subseteq U_2 \subseteq \dots \subseteq N$.

So, $\exists k \in \mathbb{N}$, such that $U_k = U_{k+1}$. Then $U_\ell = U_k$ holds $\forall \ell \geq k$.

$$U := U_k.$$

We can prove by an argument similar to the argument seen at the active nonterminals, that $X \Rightarrow_G^* \varepsilon$ holds if and only if $X \in U$.

This implies $\varepsilon \in L(G)$ if and only if $S \in U$.

Chomsky normal form

Eliminate all the rules $X \rightarrow \varepsilon$ from P ($X \in N$), these rules are called **ε -rules**. Then add the following rules.

- ▶ In the case of $A \rightarrow BC \in P, B, C \in U$ add $A \rightarrow B$ and $A \rightarrow C$ to P
- ▶ In the case of $A \rightarrow BC \in P, B \in U, C \notin U$ add $A \rightarrow C$ to P
- ▶ In the case of $A \rightarrow BC \in P, B \notin U, C \in U$ add $A \rightarrow B$ to P

Let G_1 be the grammar after the transformation. Then $L(G_1) \subseteq L(G) \setminus \{\varepsilon\}$ holds, since application of a new rule corresponds to application of an original rule followed by a derivation $B \Rightarrow_G^* \varepsilon$, $C \Rightarrow_G^* \varepsilon$, or both, respectively.

On the other hand, consider a derivation $S \Rightarrow_G^* u$ and $u \neq \varepsilon$. Then $S \Rightarrow_{G'}^* u$ holds, since application of any ε -rule can be avoided by the new rules. This implies $L(G) \setminus \{\varepsilon\} \subseteq L(G_1)$ proving $L(G) \setminus \{\varepsilon\} = L(G_1)$

Chomsky normal form

Case a) $S \notin U$

In the case $S \notin U$ G_1 is equivalent to G , RER condition holds, since G_1 has no ε -rules at all.

Case b) $S \in U$

In the case of $S \in U$ let G'_1 be the following modification of G_1 . Add a new nonterminal S' to the set of nonterminals of G_1 . Let S' be the initial symbol of G'_1 . Production rules of G'_1 contains all rules of G_1 and the rules $S' \rightarrow \varepsilon$ and $S' \rightarrow S$.

Then $L(G'_1) = L(G)$ holds. The only ε -rule of G'_1 is $S' \rightarrow \varepsilon$ and S' does not appear on any right hand sides of the rules. Sy G'_1 is of the appropriate form.

Exception: In the case $S \in U$, and S does not appear on any right hand sides introduction of the new symbol S' is not needed. It is enough to add the rule $S \rightarrow \varepsilon$ to the production rules of G_1 .

Chomsky normal form

Step 4 Eliminating chain rules

Or only task left is to eliminate the **chain rules** $X \rightarrow Y$ ($X, Y \in N$).
Let P_0 denote the set of chain rules of P .

For every $A \in N$ let $H(A) := \{B \in N \mid A \Rightarrow^* B\}$.

Then the rules of a chain rule free grammar G' equivalent to G is as follows.

$P' := \{A \rightarrow w \mid \exists B \in H(A), \text{ with } B \rightarrow w \in P\} \setminus P_0$.

$L(G') \subseteq L(G)$, since all applications of a new rule $A \rightarrow w$ corresponds to an application of a sequence of chain rules $A \Rightarrow^* B$ followed by an original rule $B \rightarrow w$ ($B \in H(A)$).

On the other hand $L(G) \subseteq L(G')$, since applications of chain rules are not needed after the introduction of the new rules.

Chomsky normal form

For the algorithmic production of $H(A)$ let us define the sets $H_i(A)$ ($i \geq 0$) recursively as follows.

$$H_0(A) := \{A\}$$

$$H_{i+1}(A) := H_i(A) \cup \{B \in N \mid \exists C \in H_i(A) \text{ with } C \rightarrow B \in P\}.$$

$$H_0(A) \subseteq H_1(A) \subseteq \dots \subseteq H_k(A) \subseteq \dots \subseteq N$$

$$k := \min\{0 \leq i \leq n - 1 \mid H_i(A) = H_{i+1}(A)\}.$$

$$H(A) := H_k(A).$$

Then, by a similar argument seen at the discussion of reachable nonterminals it can be proved, that $H_k(A)$ is the set of nonterminals reachable from A . □

Chomsky normal form – example

Example: Bring the following grammar to Chomsky normal form.
(capital letters are nonterminals, S is the initial symbol)

$$S \rightarrow AB$$

$$A \rightarrow aAa \mid C$$

$$B \rightarrow bBb \mid C$$

$$C \rightarrow Cabc \mid b \mid \varepsilon$$

Solution:

Step 1 (Eliminating rules with nonsolitary terminals):

$$S \rightarrow AB$$

$$A \rightarrow DAD \mid C$$

$$B \rightarrow EBE \mid C$$

$$C \rightarrow CDEF \mid b \mid \varepsilon$$

$$D \rightarrow a$$

$$E \rightarrow b$$

$$F \rightarrow c$$

Chomsky normal form – example

Step 2 (Length reduction):

$$\begin{array}{l} S \rightarrow AB \\ A \rightarrow DAD \mid C \\ B \rightarrow EBE \mid C \\ C \rightarrow CDEF \mid b \mid \varepsilon \\ D \rightarrow a \\ E \rightarrow b \\ F \rightarrow c \end{array} \quad \Longrightarrow \quad \begin{array}{l} S \rightarrow AB \\ A \rightarrow DZ_1 \mid C \\ B \rightarrow EZ_2 \mid C \\ C \rightarrow CZ_3 \mid b \mid \varepsilon \\ D \rightarrow a \\ E \rightarrow b \\ F \rightarrow c \\ Z_1 \rightarrow AD \\ Z_2 \rightarrow BE \\ Z_3 \rightarrow DZ_4 \\ Z_4 \rightarrow EF \end{array}$$

Chomsky normal form – example

Step 3 (Eliminating ε -rules):

$$U_1 = \{C\}, U_2 = \{C, A, B\}, U_3 = U_4 = \{S, A, B, C\} = U.$$

$$S \rightarrow AB$$

$$A \rightarrow DZ_1 \mid C$$

$$B \rightarrow EZ_2 \mid C$$

$$C \rightarrow CZ_3 \mid b \mid \varepsilon$$

$$D \rightarrow a$$

$$E \rightarrow b$$

$$F \rightarrow c$$

$$Z_1 \rightarrow AD$$

$$Z_2 \rightarrow BE$$

$$Z_3 \rightarrow DZ_4$$

$$Z_4 \rightarrow EF$$

$\implies$

$$S \rightarrow AB \mid A \mid B \mid \varepsilon$$

$$A \rightarrow DZ_1 \mid C$$

$$B \rightarrow EZ_2 \mid C$$

$$C \rightarrow CZ_3 \mid b \mid Z_3$$

$$D \rightarrow a$$

$$E \rightarrow b$$

$$F \rightarrow c$$

$$Z_1 \rightarrow AD \mid D$$

$$Z_2 \rightarrow BE \mid E$$

$$Z_3 \rightarrow DZ_4$$

$$Z_4 \rightarrow EF$$

Remark: $S \in U$, so we are in case b). S does not appear on any right hand sides, so no new initial symbol is introduced.

Chomsky normal form – example

Step 4 (Eliminating chain rules):

$$H_0(S) = \{S\}, H_1(S) = \{S, A, B\}, H_2(S) = \{S, A, B, C\}, H_3(S) = \{S, A, B, C, Z_3\} = H(S)$$

Similarly $H(A) = \{A, C, Z_3\}$, $H(B) = \{B, C, Z_3\}$, $H(C) = \{C, Z_3\}$, $H(Z_1) = \{D, Z_1\}$, $H(Z_2) = \{E, Z_2\}$, the other sets are trivial.

$$S \rightarrow AB \mid A \mid B \mid \varepsilon$$

$$A \rightarrow DZ_1 \mid C$$

$$B \rightarrow EZ_2 \mid C$$

$$C \rightarrow CZ_3 \mid b \mid Z_3$$

$$D \rightarrow a$$

$$E \rightarrow b$$

$$F \rightarrow c$$

$$Z_1 \rightarrow AD \mid D$$

$$Z_2 \rightarrow BE \mid E$$

$$Z_3 \rightarrow DZ_4$$

$$Z_4 \rightarrow EF$$

$\Rightarrow$

$$S \rightarrow AB \mid DZ_1 \mid CZ_3 \mid b \mid DZ_4 \mid EZ_2 \mid \varepsilon$$

$$A \rightarrow DZ_1 \mid CZ_3 \mid b \mid DZ_4$$

$$B \rightarrow EZ_2 \mid CZ_3 \mid b \mid DZ_4$$

$$C \rightarrow CZ_3 \mid b \mid DZ_4$$

$$D \rightarrow a$$

$$E \rightarrow b$$

$$F \rightarrow c$$

$$Z_1 \rightarrow AD \mid a$$

$$Z_2 \rightarrow BE \mid b$$

$$Z_3 \rightarrow DZ_4$$

$$Z_4 \rightarrow EF$$

Chomsky normal form – efficiency of the algorithm

Let $G = (N, \Sigma, P, S)$ be a context-free grammar. For a rule $p : X \rightarrow Y_1 \cdots Y_m \in P$ let $|p| := m + 1$, the sum of the lengths of the left and right hand sides of p . Let $|G| := \sum_{p \in P} |p|$ be the **size** of the grammar G .

Efficiency of the algorithm:

Our algorithm constructs a grammar in Chomsky normal form of size asymptotically $|G|^2$ in asymptotically $|G|^2$ steps.

Remark: All the steps are of linear magnitude except for eliminating chain rules.

Remark: It is possible to eliminate ε -rules for CF grammars having long rules, but this may result an exponential growth in the number of rules. E.g., consider the rule $A \rightarrow BCDE$ and assume that $U = \{C, D\}$. Then this rule is replaced by the rules $A \rightarrow BCDE \mid BCD \mid CDE \mid CD$.

Membership problem of context-free grammars

Claim: Let $G = (N, \Sigma, P, S)$ be a context-free grammar and $u \in \Sigma^*$ be a word. It is algorithmically decidable whether $u \in L(G)$ holds or not.

Proof: According to our previous theorem there exist a grammar G' in Chomsky normal form equivalent with G . This grammar can be produced algorithmically. Let $n = |u|$.

If $n = 0$ then $u \in L(G) \Leftrightarrow S \rightarrow \varepsilon \in P$.

It is easy to see by induction, that any derivation of u of length $n \geq 1$ consists of exactly $n - 1$ applications of an ' $A \rightarrow BC$ type' rule and n applications of an ' $A \rightarrow a$ ' rule ($A, B, C \in N, a \in \Sigma$).

The number of such derivations is finite and can be checked by an exhaustive search.

Membership problem of context-free grammars

The previous argument gives an exponential bound for the efficiency of the membership problem.

We can make a search for the derivations of u by using a 'bottom-up' 'dynamic programming' strategy. This approach gives a polynomial upper bound.

The algorithm is due to John Cocke, Daniel Younger and Tadao Kasami (CYK algorithm) and solves the membership problem in $O(|G| \cdot n^3)$ steps if the grammar is in Chomsky normal form.

For general grammars we have an upper bound $O(|G|^2 \cdot n^3)$ for the efficiency.

Membership problem of context-free grammars

CYK algorithm

Cocke-Younger-Kasami (CYK) algorithm

Input: A context-free grammar $G = (N, \Sigma, P, S)$ in **Chomsky normal form** and a word $u \in \Sigma^*$.

Output: $u \stackrel{?}{\in} L(G)$

Let $u = t_1 \dots t_n$, $t_i \in \Sigma$, $n \geq 1$. Let A_i be the left and β_i be the right hand side of $P_i \in P$ ($1 \leq i \leq |P|$, $A_i \in N$, $\beta_i \in \Sigma \cup N^2$.)

CYK algorithm defines sets $H_{i,j}$, $1 \leq i \leq j \leq n$ recursively by incrementing $(j-i)$.

$$H_{i,i} := \{A_j \mid \beta_j = t_i\}$$

$$H_{i,j} := \{A_k \mid \beta_k \in \bigcup_{h=i}^{j-1} H_{i,h} H_{h+1,j}\} \quad (i < j)$$

$$u \in L(G) \iff S \in H_{1,n}.$$

Soundness of CYK algorithm

To prove the soundness of the algorithm it is enough to see, that $H_{i,j} = \{X \in N \mid X \Rightarrow_G^* t_i \cdots t_j\}$ holds for every $1 \leq i \leq j \leq n$.

We prove this claim by induction of $(j - i)$.

By definition t_i can be derived exactly from the members $H_{i,i}$ proving the case $j - i = 0$.

Let $1 \leq i < j \leq n$. Assume, that statement holds for all pairs of integers $1 \leq k \leq \ell \leq n$, where $\ell - k < j - i$.

Consider a derivation $X \Rightarrow^* t_i \cdots t_j$. Since $j - i \geq 1$ first step of the derivation should be $X \Rightarrow YZ$ for some $Y, Z \in N$. So, there exists an index $i \leq h < j$ with $Y \Rightarrow^* t_i \cdots t_h$ and $Z \Rightarrow^* t_{h+1} \cdots t_j$. Since $h - i < j - i$ and $j - (h + 1) < j - i$ hold we have $Y \in H_{i,h}$ and $Z \in H_{h+1,j}$ by induction implying $X \in H_{i,j}$.

Let $X \in H_{i,j}$ ($j > i$). Then there should be an index $i \leq h < j$ and $Y, Z \in N$ with $Y \in H_{i,h}$, $Z \in H_{h+1,j}$ and $X \rightarrow YZ \in P$. So we have $Y \Rightarrow^* t_i \cdots t_h$ and $Z \Rightarrow^* t_{h+1} \cdots t_j$ by induction. Then we have $X \Rightarrow YZ \Rightarrow^* t_i \cdots t_h Z \Rightarrow^* t_i \cdots t_j$.

CYK algorithm

Example: Decide by CYK algorithm whether $aabbcc$ can be derived in $G = (\{S, A, B, C, U, V, W, X, Y, Z\}, \{a, b, c\}, P, S)$ where P is as follows.

$$S \rightarrow AB \mid BC$$
$$A \rightarrow XA \mid a$$
$$X \rightarrow a$$
$$C \rightarrow YC \mid c$$
$$Y \rightarrow C$$
$$B \rightarrow UV \mid VW \mid XS$$
$$U \rightarrow XX$$
$$W \rightarrow YY \mid XS$$
$$V \rightarrow ZZ$$
$$Z \rightarrow b$$
$$H_{1,n}$$
$$H_{1,n-1}$$
$$H_{2,n}$$

•

•

•

$$H_{1,3}$$
 $H_{2,4}$
$$H_{1,2}$$
 $H_{2,3}$

• • •

$$H_{n-1,n}$$
 $H_{1,1}$ $H_{2,2}$

• • •

$$H_{n,n}$$
 t_1 t_2

• • •

 t_n

CYK algorithm

$S \rightarrow AB \mid BC$

$A \rightarrow XA \mid a$

$X \rightarrow a$

$C \rightarrow YC \mid c$

$Y \rightarrow c$

$B \rightarrow UV \mid VW \mid XS$

$U \rightarrow XX$

$W \rightarrow YY \mid XS$

$V \rightarrow ZZ$

$Z \rightarrow b$

$\{Y, C\}$

$a \quad a \quad b \quad b \quad c \quad c$

CYK algorithm

$S \rightarrow AB \mid BC$

$A \rightarrow \text{XA} \mid a$

$X \rightarrow a$

$C \rightarrow YC \mid c$

$Y \rightarrow c$

$B \rightarrow UV \mid VW \mid XS$

$U \rightarrow \text{XX}$

$W \rightarrow YY \mid XS$

$V \rightarrow ZZ$

$Z \rightarrow b$

$\{A, U\}$

$\{A, X\}$	$\{A, X\}$	$\{Z\}$	$\{Z\}$	$\{Y, C\}$	$\{Y, C\}$
------------	------------	---------	---------	------------	------------

a	a	b	b	c	c
-----	-----	-----	-----	-----	-----

$\{A, X\}\{A, X\} = \{AA, AX, XA, XX\}$

CYK algorithm

$S \rightarrow AB \mid BC$

$A \rightarrow XA \mid a$

$X \rightarrow a$

$C \rightarrow YC \mid c$

$Y \rightarrow c$

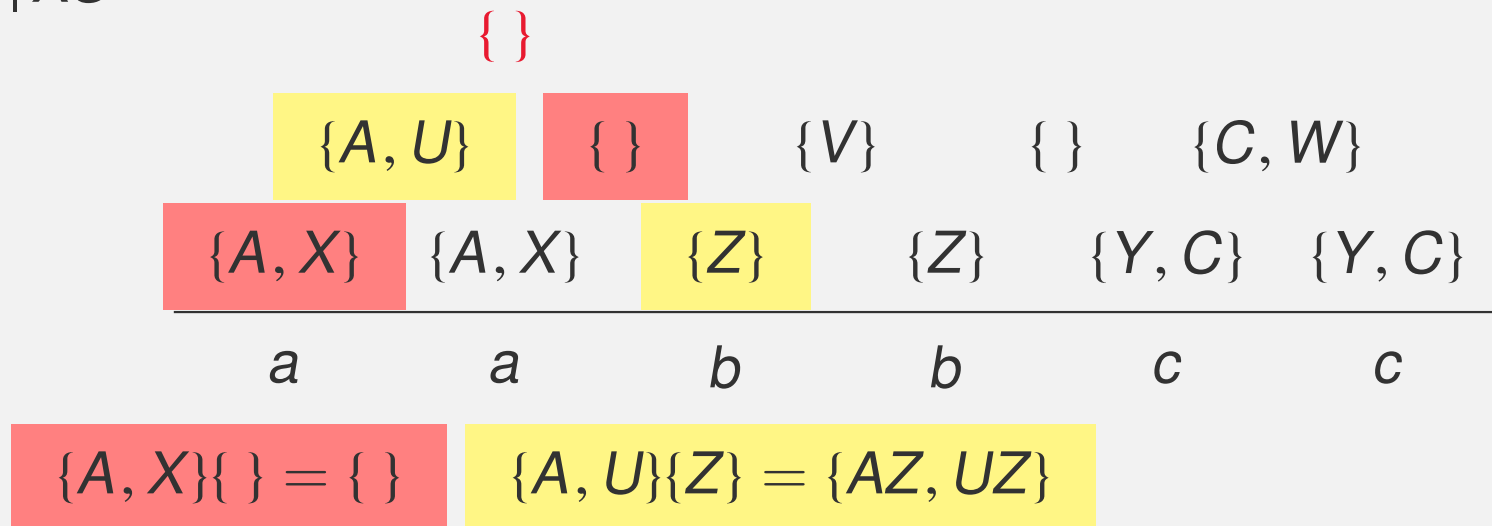
$B \rightarrow UV \mid VW \mid XS$

$U \rightarrow XX$

$W \rightarrow YY \mid XS$

$V \rightarrow ZZ$

$Z \rightarrow b$



CYK algorithm

$S \rightarrow AB \mid BC$

$A \rightarrow XA \mid a$

$X \rightarrow a$

$C \rightarrow YC \mid c$

$Y \rightarrow c$

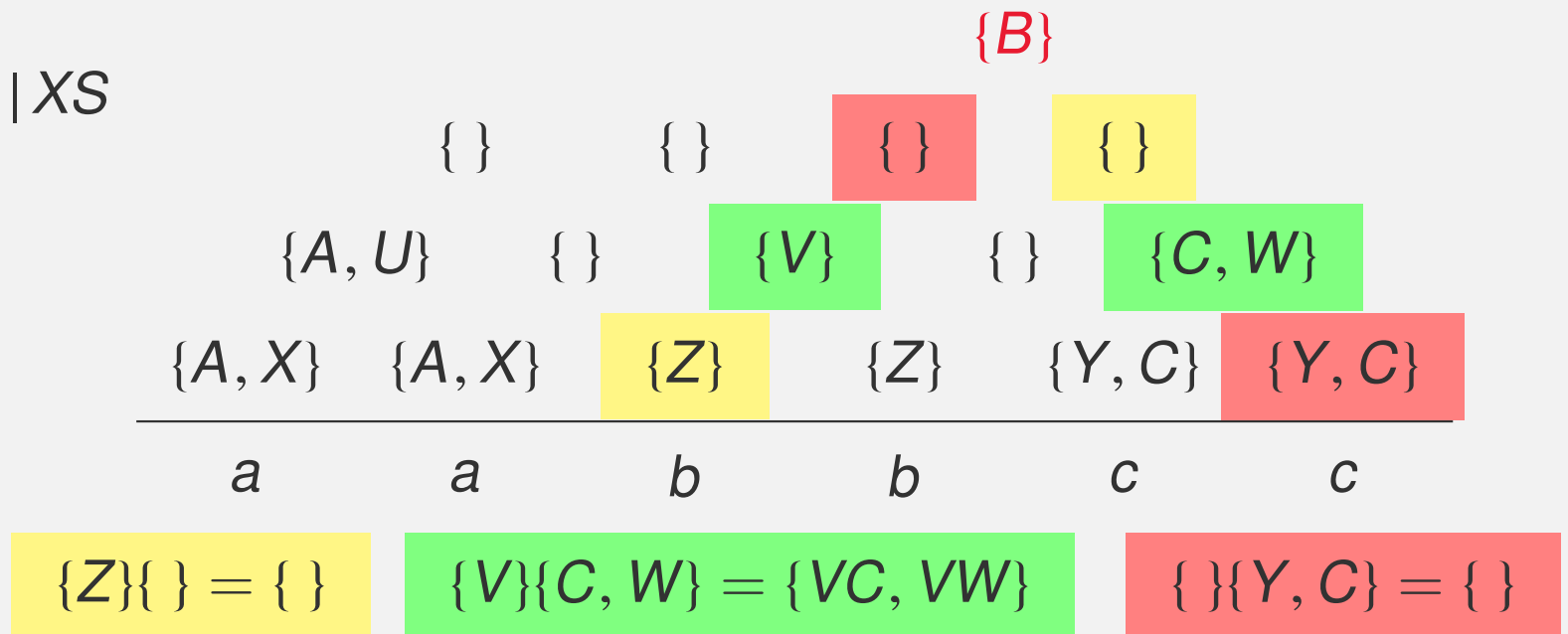
$B \rightarrow UV \mid \text{VW} \mid XS$

$U \rightarrow XX$

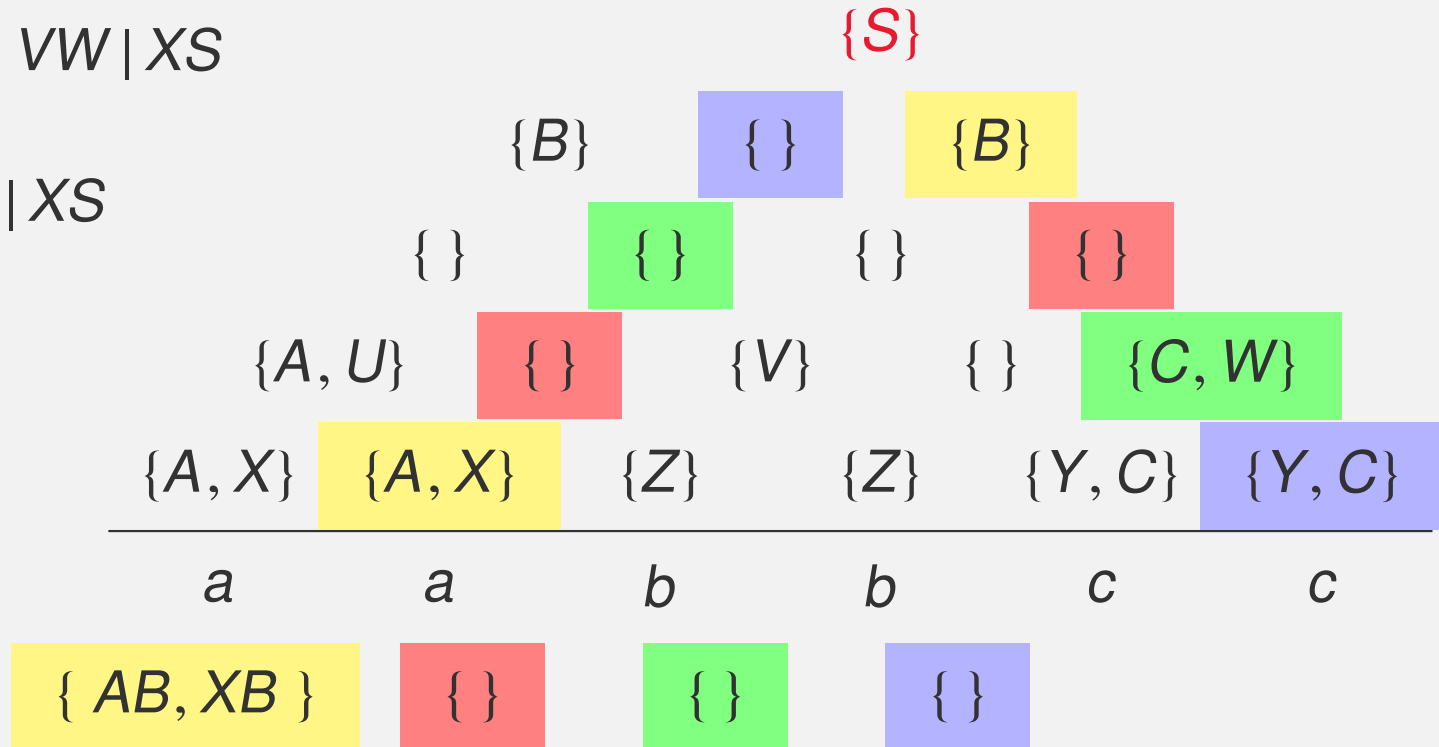
$W \rightarrow YY \mid XS$

$V \rightarrow ZZ$

$Z \rightarrow b$



CYK algorithm

$$S \rightarrow AB \mid BC$$
$$A \rightarrow XA \mid a$$
$$X \rightarrow a$$
$$C \rightarrow YC \mid c$$
$$Y \rightarrow C$$
$$B \rightarrow UV \mid VW \mid XS$$
$$U \rightarrow XX$$
$$W \rightarrow YY \mid XS$$
$$V \rightarrow ZZ$$
$$Z \rightarrow b$$


CYK algorithm

$S \rightarrow AB \mid BC$

$A \rightarrow XA \mid a$

$X \rightarrow a$

$C \rightarrow YC \mid c$

$Y \rightarrow c$

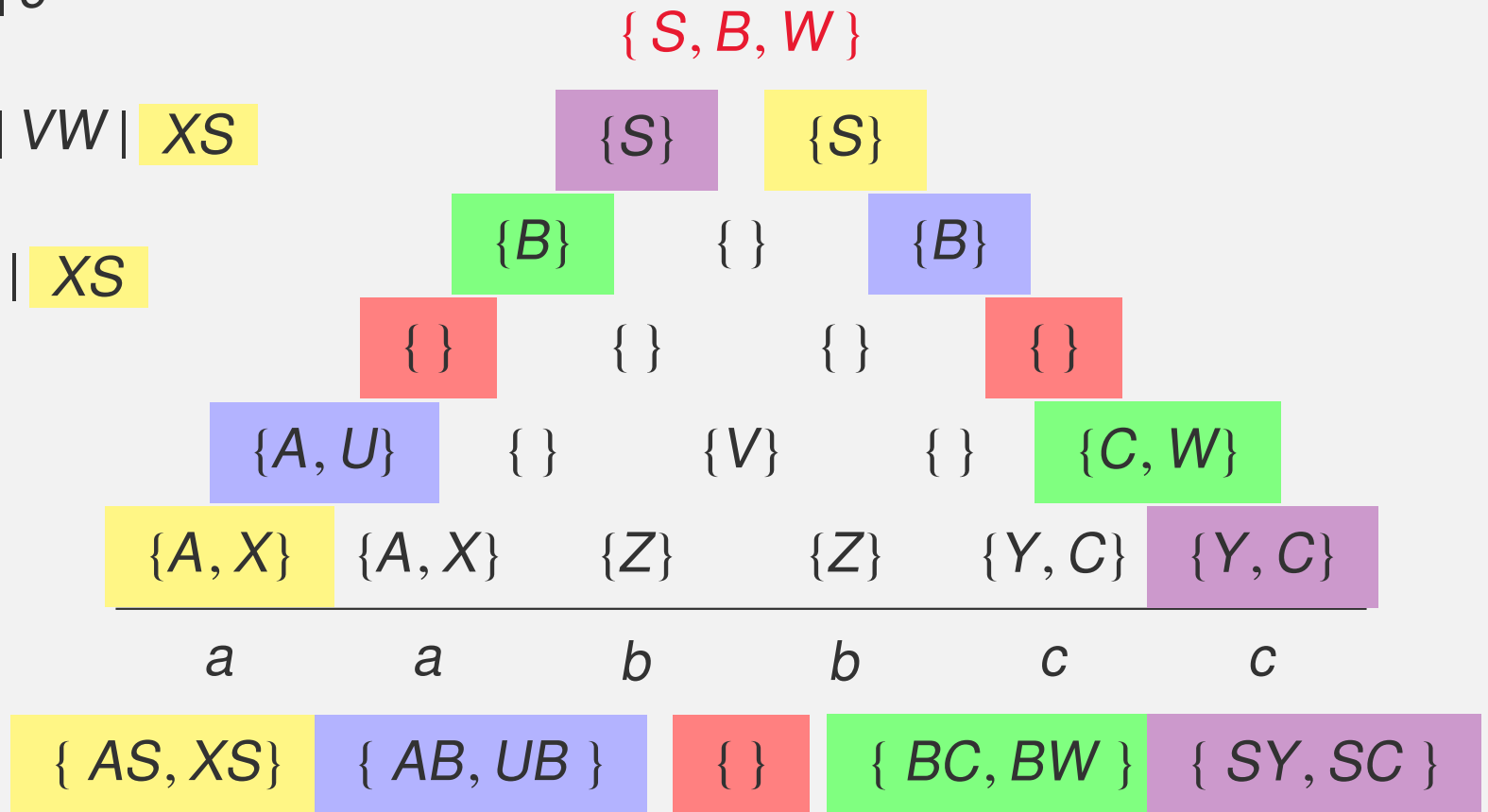
$B \rightarrow UV \mid VW \mid XS$

$U \rightarrow XX$

$W \rightarrow YY \mid XS$

$V \rightarrow ZZ$

$Z \rightarrow b$



CYK algorithm

$$S \rightarrow AB \mid BC$$
$$A \rightarrow XA \mid a$$
$$X \rightarrow a$$
$$C \rightarrow YC \mid c$$
$$Y \rightarrow C$$
$$B \rightarrow UV \mid VW \mid XS$$
$$U \rightarrow XX$$
$$W \rightarrow YY \mid XS$$
$$V \rightarrow ZZ$$
$$Z \rightarrow b$$
 $\{S, B, W\}$ $\{S\}$ $\{S\}$ $\{B\}$ $\{ \}$
$$\{B\}$$
 $\{ \}$ $\{ \}$ $\{\}$ $\{ \}$
$$\{A, U\}$$
 $\{\}$ $\{V\}$ $\{$ $\{C, W\}$
$$\{A, X\}$$
$$\{A, X\}$$
 $\{Z\}$
$$\{Z\}$$
 $\{Y, C\}$ $\{Y, C\}$

a

a

b

b

C

C

Since $S \in H_{1,6}$, we have $aabbcc \in L(G)$.

Derivations can be obtained by keeping track of the pairs that make nonterminals members of the sets $H_{i,j}$.

$$S \Rightarrow AB \Rightarrow XAB \Rightarrow XAVW \Rightarrow XAZZW \Rightarrow XAZZYY \Rightarrow^* aabbcc.$$